

AnatoBind-MRI Architecture and loss functions

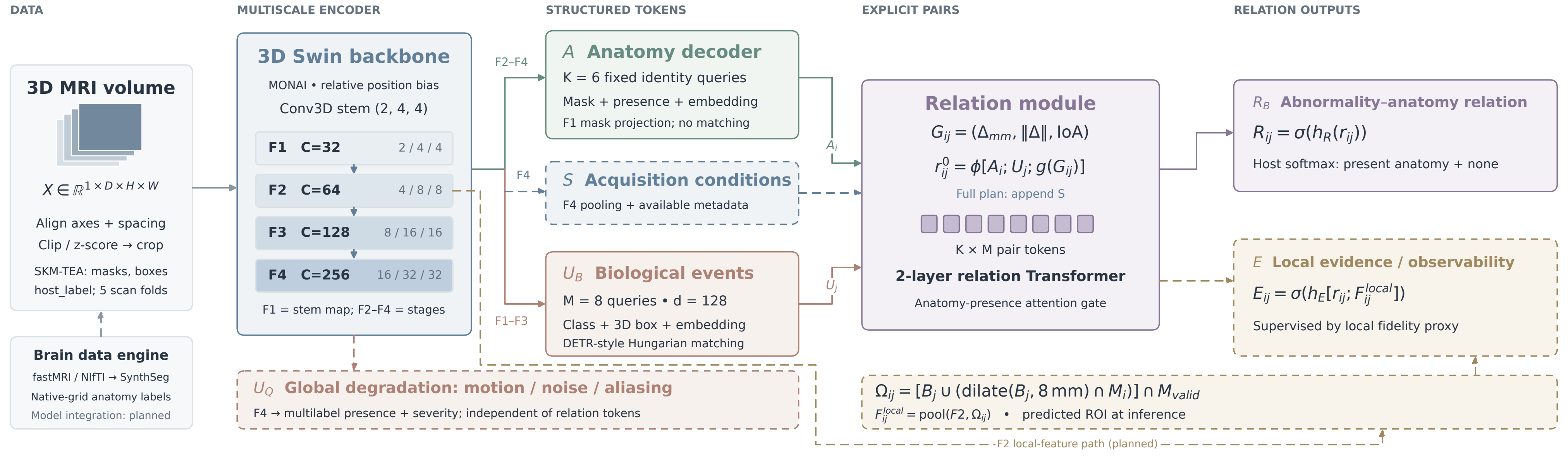
foundation_model / code snapshot f812043 / research plan v2.1 / 08 Sep 2026

Implemented core

Planned addition

Dotted: supervision

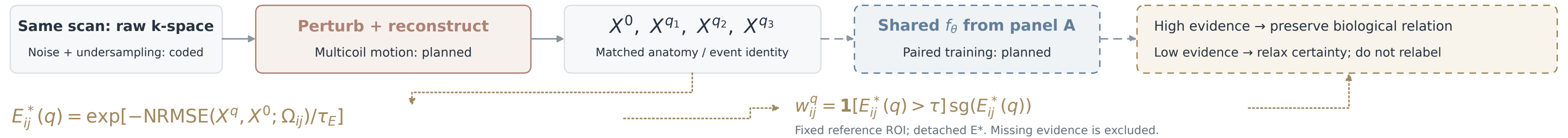
A MRI → entities / events → relations → local evidence



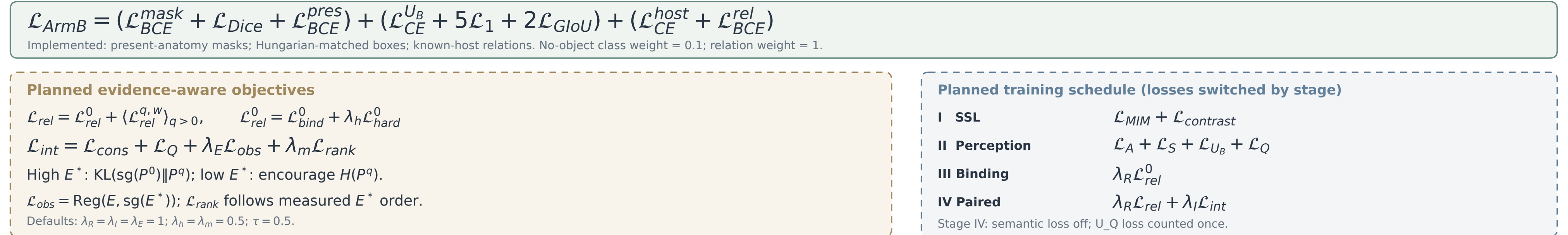
CURRENT LIMIT: GT masks + matched GT boxes + GT presence feed relation geometry/gates. Host accuracy is NOT M1 evidence.

Full plan: cross-organ ontology, d=256 / M=20, wider Swin, physical 3D RoPE, valid-voxel + event gates, S / U_Q / E, and staged paired training.

B Controlled k-space views and external evidence supervision



C Loss functions: executable objective and full-plan extensions



Interpretation: R predicts relation membership; E estimates local evidence via a fidelity proxy. Empty ROI / missed event → no E score; report coverage.

Source: anatobind/{data_engine,model,train} + RESEARCH_PLAN.md §§4–8. Proposed observability and generalization remain unvalidated.